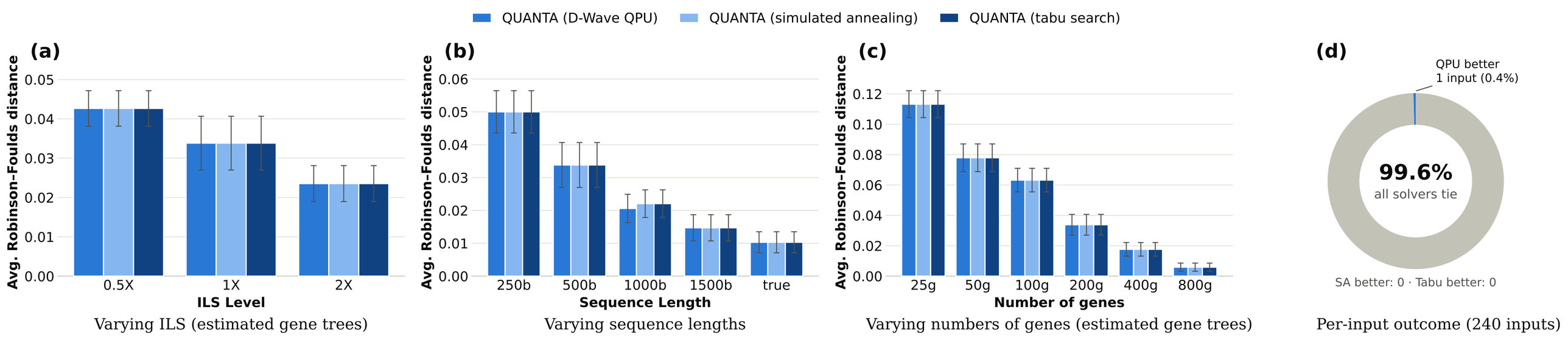
QUANTA (D-Wave QPU)
QUANTA (simulated annealing)
QUANTA (tabu search)
(a)
Avg. Robinson–Foulds distance
0.05
0.04
0.03
0.02
0.01
0.00
0.5X
1X
2X
ILS Level
Varying ILS (estimated gene trees)
(b)
Avg. Robinson–Foulds distance
0.06
0.05
0.04
0.03
0.02
0.01
0.00
250b
500b
1000b
1500b
true
Sequence Length
Varying sequence lengths
(c)
Avg. Robinson–Foulds distance
0.12
0.10
0.08
0.06
0.04
0.02
0.00
25g
50g
100g
200g
400g
800g
Number of genes
Varying numbers of genes (estimated gene trees)
(d)
QPU better
1 input (0.4%)
99.6%
all solvers tie
SA better: 0 · Tabu better: 0
Per-input outcome (240 inputs)